# Weston Pangle

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## EDUCATION

**University Of Michigan, Ann Arbor, MI** *August 2021 – May 2025*

*B.S.E. in Computer Science Engineering*

**Relevant Coursework:** DS&A, Computer Organization, Software Engineering, Web Systems, Operating Systems (6 credit), Computer Networks, Game Design & Dev, Intro Machine Learning, Distributed Systems, Database Management Systems, Computer Security

**Languages and Technologies:** Proficient - C/C++, Go, Git, GitHub, Python | Intermediate – JIRA, Unity, SQL| Familiar - AWS, CSS, Docker, HTML, Java, JavaScript, MongoDB, Rust

## LEADERSHIP AND EXPERIENCE

**Founding Engineer:** <u>**Pick It Up!**</u> **–** *Driver's Seat Studios* *October 2024 – Current*

- Spearheaded the design of dynamic, immersive level layouts and engaging gameplay systems that balanced challenge and pacing, directly contributing to the team's 1st place finish out of 14 competing games at a university game showcase event
- Generated strong early traction with 300+ web downloads by enhancing player engagement through polished mechanics and immersive environmental storytelling, while actively pursuing publishing deals to expand distribution and enhance gameplay
- Integrated and optimized high-quality assets, including sophisticated particle effects, custom Shader Graph cel shading, and interactive environmental systems to elevate visual fidelity and maintain robust performance across multiple systems

**Orientation Leader –** *University of Michigan* *April 2024 – Current*

- Coordinated daily orientations for 1,000–1,500 attendees by organizing schedules, managing logistics, and delivering presentations, demonstrating strong organizational skills, public speaking ability, and clear communication of information
- Facilitated small group activities, breakout discussions, and Q&A sessions, collaborating with team members to ensure smooth operations and create a comfortable, welcoming environment that encouraged student participation and engagement

**ML Engineer –** *Michigan Data Science Team: University of Michigan* *September 2022 – January 2023*

- Developed a Pokémon "auto-battler" using Python and supervised learning techniques, achieving 85% success in battle predictions and optimizing performance with a move recommendation algorithm based on Pokémon types, moves, and traits
- Collaborated with a team to design, train, and refine machine learning models for strategic battle simulations, significantly enhancing the AI's ability to predict and adapt to battle scenarios, leading to improved gameplay dynamics and outcomes

## PERSONAL PROJECTS

**Simple Rust Operating System | Rust** May 2025 – Current

- Engineered a bare-metal 64-bit x86_64 Rust kernel with VGA text-mode output, leveraging the bootloader crate and a custom linker to perform low-level hardware initialization and environment setup without standard library dependencies
- Developing critical OS mechanisms including paging for virtual memory management, CPU exception handling with detailed fault recovery routines, and double fault handler to ensure system stability and controlled hardware interaction
- Implementing a thread-safe dynamic heap allocator and async runtime using the spin crate, enabling safe, concurrent memory access and task execution within the #![no_std] context, while maintaining precise control over synchronization & interrupt safety

**Sharded Key-Value Service w/ Paxos Groups | Distributed Systems | Go** *April 2025*

- Engineered a distributed, fault-tolerant key/value store with Paxos-based replication and dynamic sharding across replica groups, ensuring high availability and consistency even during group joins, failures, and network partitions
- Designed and implemented a Shard Master service to handle consistent, serialized reconfiguration, enabling smooth and reliable shard reassignment during group joins and failures, and maintaining system stability under high load
- Achieved linearizable client operations with at-most-once semantics, ensuring minimal data loss and improved system performance during stateful shard migration, while effectively managing concurrent reconfigurations and reducing downtime

**Network File System | Advanced Operating Systems | C++** *April 2024*

- Engineered a multithreaded network file system, utilizing a manager-worker threading model to optimize handling of concurrent file operations, significantly boosting system throughput and reducing latency in high-load scenarios
- Designed and implemented thread-safe file and directory operations with mutexes and upgradeable locks, ensuring strong data integrity while minimizing performance bottlenecks during simultaneous access and improving system responsiveness
- Conducted extensive testing and validation of system performance across various operational states, ensuring robust compatibility with both initialized and new file systems while maintaining stability and reliability under heavy client load

## ADDITIONAL

**More Projects:** HTTP Proxy w/ DNS, Thread Library, Reliable Trans. Protocol, Instagram Clone, Search Engine, Metroid Remake

**Interests:** Playing Video Games, Collecting Cologne, Watching Basketball, Weightlifting, Photography